Terrace Oasis: a city that grows its own shade beside the first fields

Every mandatory requirement and agent task

Twelve mandatory requirements (M1-M12), phase one at full occupancy

	Mandatory requirement	This scheme	Evidence	Answered in
M1	Hourly carbon-free energy match	100 %	assumed	§3. Energy and compute
M2	Peak IT load in phase one	40 MW	declared	§3. Energy and compute
M3	Annual water demand, by use	3,599,292 m3/yr	assumed	§4. Water
M4	Cooling water intensity (WUE)	0.20 L/kWh	assumed	§4. Water
M5	Response to a 30% water shortfall	stated in the text	-	§4. Water
M6	Human-machine at-grade interfaces	30	computed	§5. Human and machine
M7	Machine traffic rule	combined	declared	§5. Human and machine
M8	Public audit mechanism	stated in the text	-	§8. Governance and data
M9	Control and succession of the […]	stated in the text	-	§8. Governance and data
M10	Triggers for phases two and three	stated in the text	-	§9. Phasing
M11	Irreversible decisions	stated in the text	-	§9. Phasing
M12	Jobs and residents reconcile	7,891 jobs · 41,340 persons […]	declared	§6. Programme and production

Eight agent tasks

	Agent task	Answered in
A1	Identity of the place	§1. Position
A2	Energy and compute	§3. Energy and compute
A3	Water in a dry place	§4. Water
A4	Human and machine	§5. Human and machine
A5	Programme and production	§6. Programme and production
A6	Urban form across the whole site	§7. Urban form
A7	Data and AI governance	§8. Governance and data
A8	Phasing in three steps	§9. Phasing

How to read this page

Values come from metrics.json with their evidence tags (figure 07); sections are the headings of proposal.md. The self-check (gate G6) confirms each answer points at a real heading; it does not judge the answer.

What each answer must state (brief/requirements.json)

M1 generation mix, storage duration, and whether the match is physical (self-supplied) or contractual
M2 the grid connection capacity you assumed (not supplied by the organiser)
M3 Phase one at full occupancy. The breakdown must add up to the total (2% tolerance); say which line panel-cleaning water sits in.
M4 cooling type: air, liquid, hybrid or dry - and the PUE that goes with it
M5 30% short means 30% less water than your phase-one annual demand (M3). Give a priority order of what stops first, and what happens in the city at each step.
M6 Count every place where a people or shared route meets a machine route on the same level: road crossings, the ends of shared streets, kerbs or loading points that people and machines share, stops of driverless shuttles on a […]
M7 where each rule applies and how it is enforced
M8 who publishes power, water, generation and compute data, how often, and who can check it independently
M9 who owns and operates it, and what happens if that operator leaves
M10 observable conditions, not dates
M11 at least three phase-one decisions that cannot be changed later, and why that is acceptable
M12 if residents far exceed jobs, who the others are and what they live on

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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Every declared figure with its evidence tag

Metric	Declared	Unit	Evidence	Basis / check
master_area_km2	36	km2	computed	recomputed 36
site_coverage_pct	2.92	%	computed	recomputed 2.92
gfa_m2	2,309,984	m2	computed	recomputed 2,309,984
far	0.06	ratio	computed	recomputed 0.06
open_space_pct	9.91	%	computed	recomputed 9.91
human_machine_interfaces	30	count	computed	recomputed 30
shaded_pedestrian_pct	89	%	computed	recomputed 89
structure_km2	per system	km2	computed	recomputed: per system
hourly_cfe_match_pct	100	%	assumed	A-07
compute_availability_pct	85	%	assumed	A-07
peak_it_load_mw	40	MW	declared	
pv_capacity_mwp	410	MWp	declared	
storage_mwh	1,000	MWh	declared	
pue	1.25	ratio	assumed	A-04
wue_l_per_kwh	0.20	L/kWh	assumed	A-04
waste_heat_reuse_pct	9	%	assumed	A-13
annual_water_demand_m3	3,599,292	m3/yr	assumed	A-01
water_domestic_m3	1,961,583	m3/yr	assumed	A-08
water_compute_m3	59,568	m3/yr	assumed	A-04
water_agriculture_m3	1,218,141	m3/yr	assumed	A-09
water_landscape_m3	300,000	m3/yr	assumed	A-09
water_industry_m3	60,000	m3/yr	assumed	A-09
water_reuse_pct	90	%	assumed	A-08
machine_traffic_rule	combined	enum	declared	
jobs_phase1	7,891	jobs	declared	
residents_phase1	41,340	persons	declared	
dwellings_phase1	17,224	units	declared	
long_term_residents	160,000	persons	declared	

Evidence tags

- computed - recomputed from your geometry
- sourced - cited in sources.json
- assumed - declared in assumptions.json
- indicative - organiser's indicative material

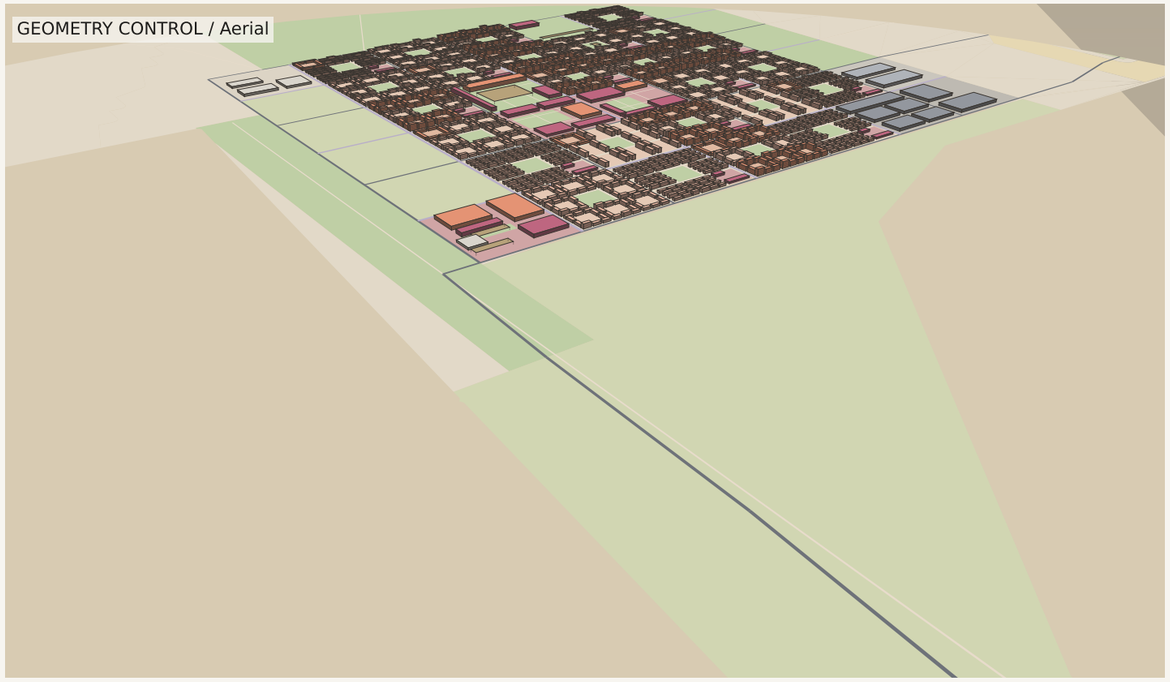
Gate G4

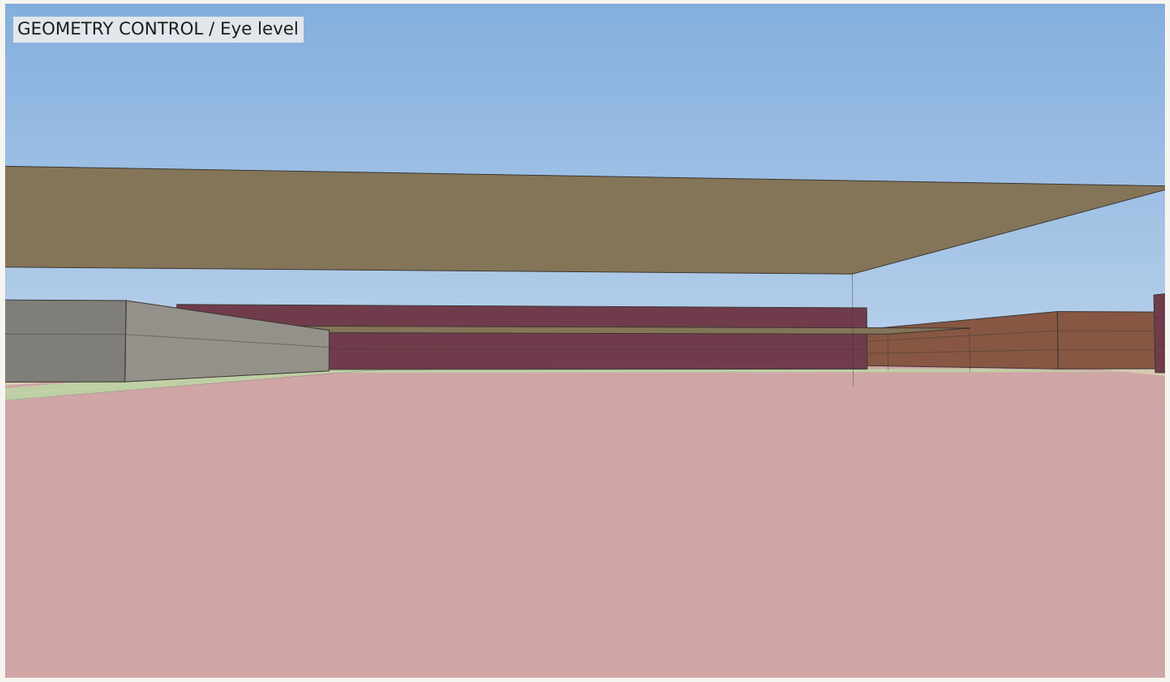
Every metric tagged computed is recomputed from geometry/ by the self-check; a declared value more than 2% away fails.

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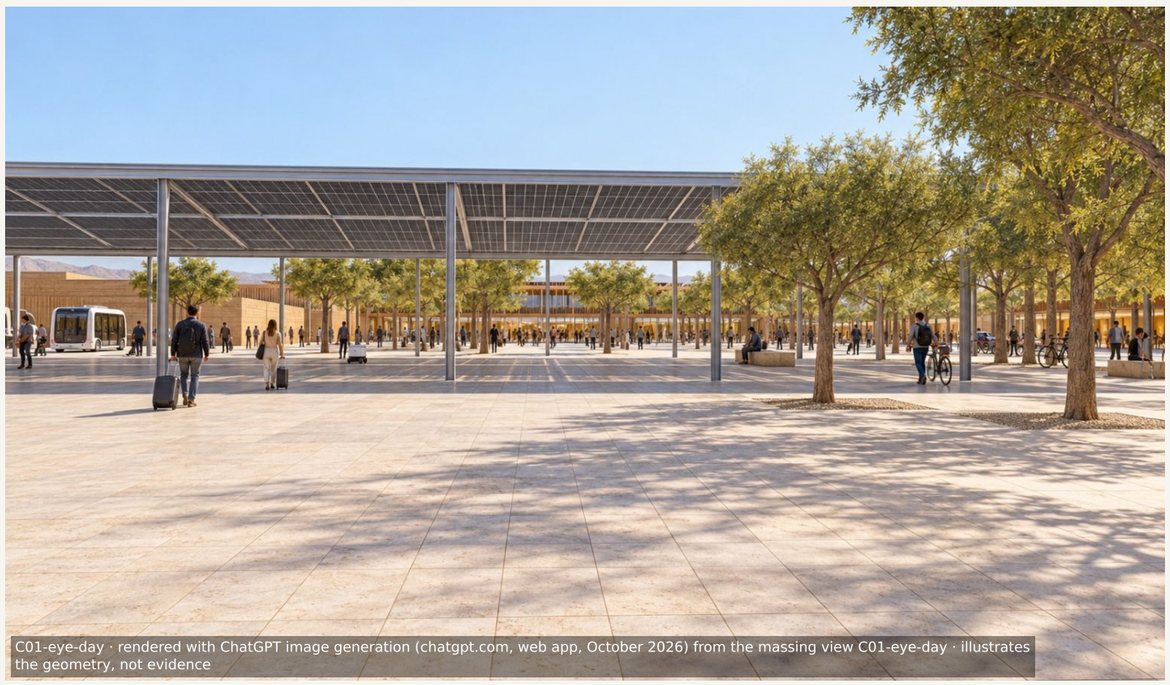
Control images remain evidence; concept images illustrate the submitted geometry







C01-aerial · rendered with ChatGPT image generation (chatgpt.com, web app, October 2026) from the massing view C01-aerial · illustrates the geometry, not evidence



C01-eye-day · rendered with ChatGPT image generation (chatgpt.com, web app, October 2026) from the massing view C01-eye-day · illustrates the geometry, not evidence

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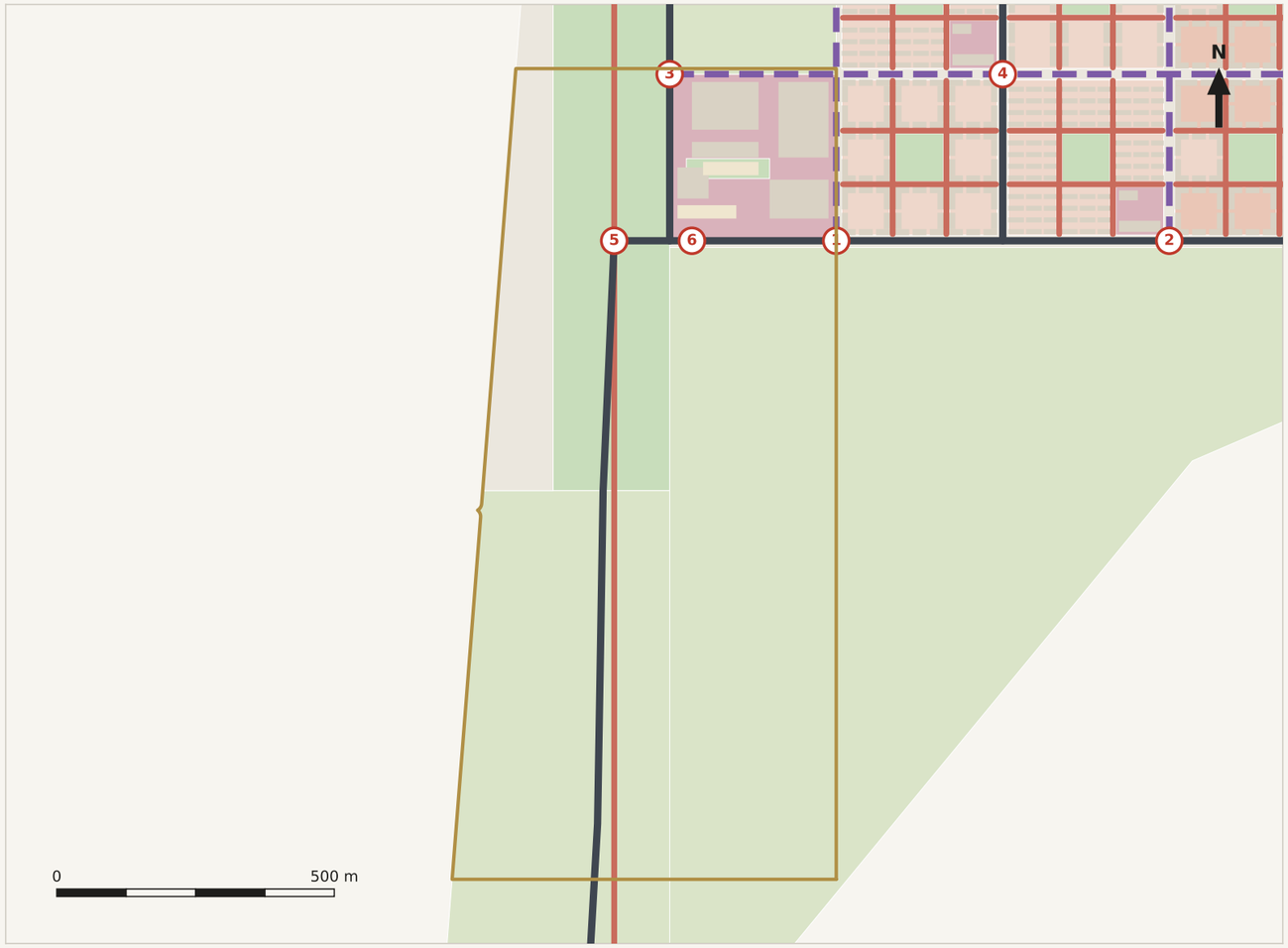
Terrace Oasis: a city that grows its own shade beside the first fields

A day here

	People	Machines
05:30	Field crews for the PV and jojoba shifts meet under the canopy; the arrival hall opens its cafe.	Panel-cleaning robots finish their night pass on the trackers; the first driverless coach leaves for San Juan.
07:30	People from San Juan and the valley step off the coach and walk in on foot; children cycle along the alameda to school.	Coaches and shuttles stop under the canopy; freight stays on the machine street to the cross-dock.
11:00	Visitors start at the arrival hall: the model of the city, guides, bikes for hire.	Delivery robots take luggage to the guesthouse; the dock rebalances bicycles and charges drones.
15:00	The hottest hours: people wait under the canopy and in the hall, and walk in the shade of the poplars.	The canopy's PV runs at its peak and charges the hub's batteries; fewer coaches run.
19:00	The market hall sells the fields' oil, nuts and vegetables; families meet arriving relatives on the square.	Robots bring crates from the agri-processing yard to the stalls along the shared street.
22:00	The last coach; the square is lit low and warm, and the sky over the fields is kept dark.	Freight moves only on the machine street; drones land at the dock for the night.

Who and what moves here: networks, interfaces, and the people, robots and shuttles of the 3D scene

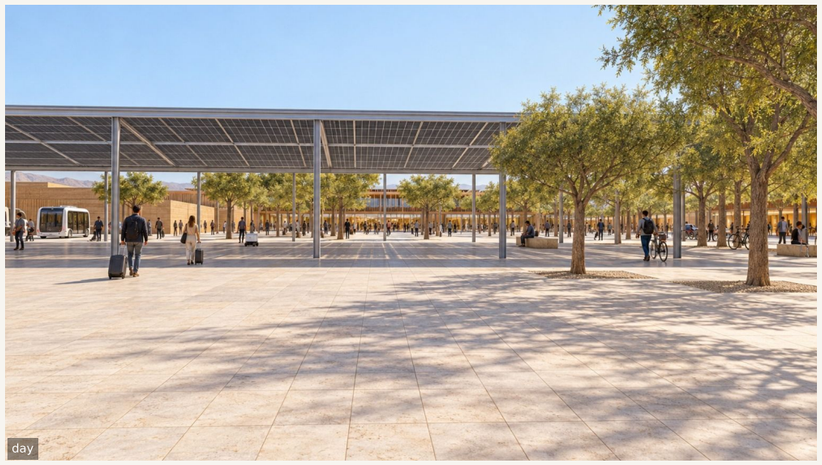
People and machines through one day



Interfaces in view

1 Raised table crossing 4
2 Raised table crossing 8
3 Raised table crossing 13
4 Raised table crossing 15
5 Raised table crossing 25
6 Driverless shuttle stop 3

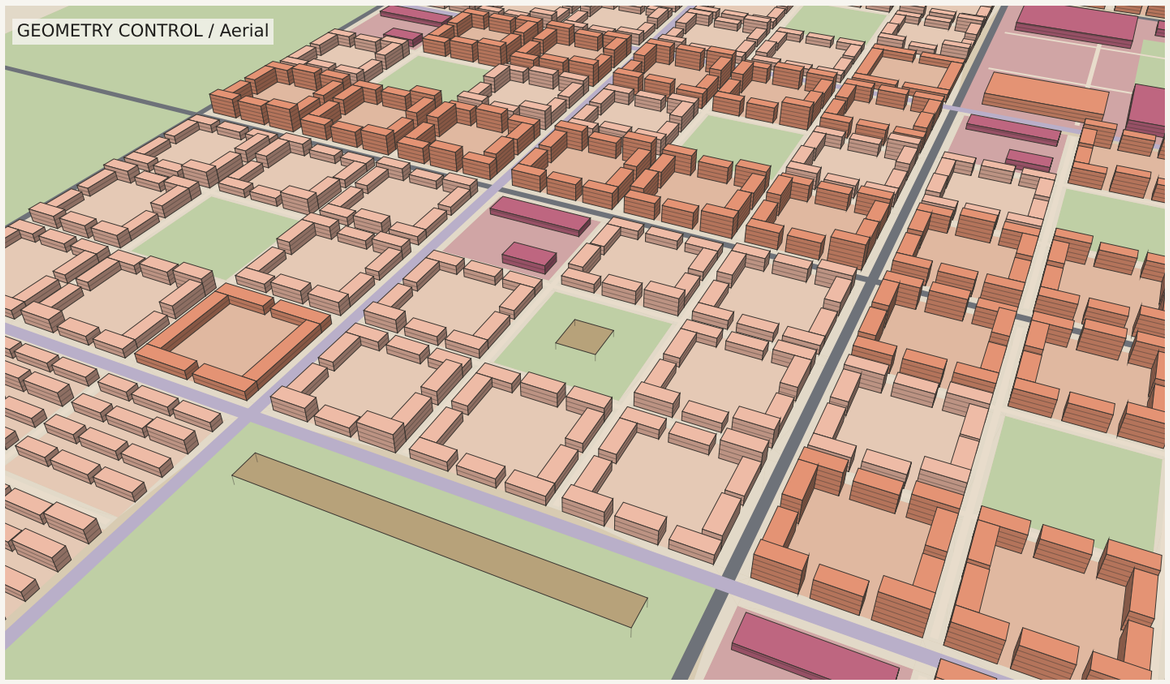
People
Machines + goods
Shared at walking pace
People (scene)
Delivery robots
Driverless shuttles
Human-machine interface

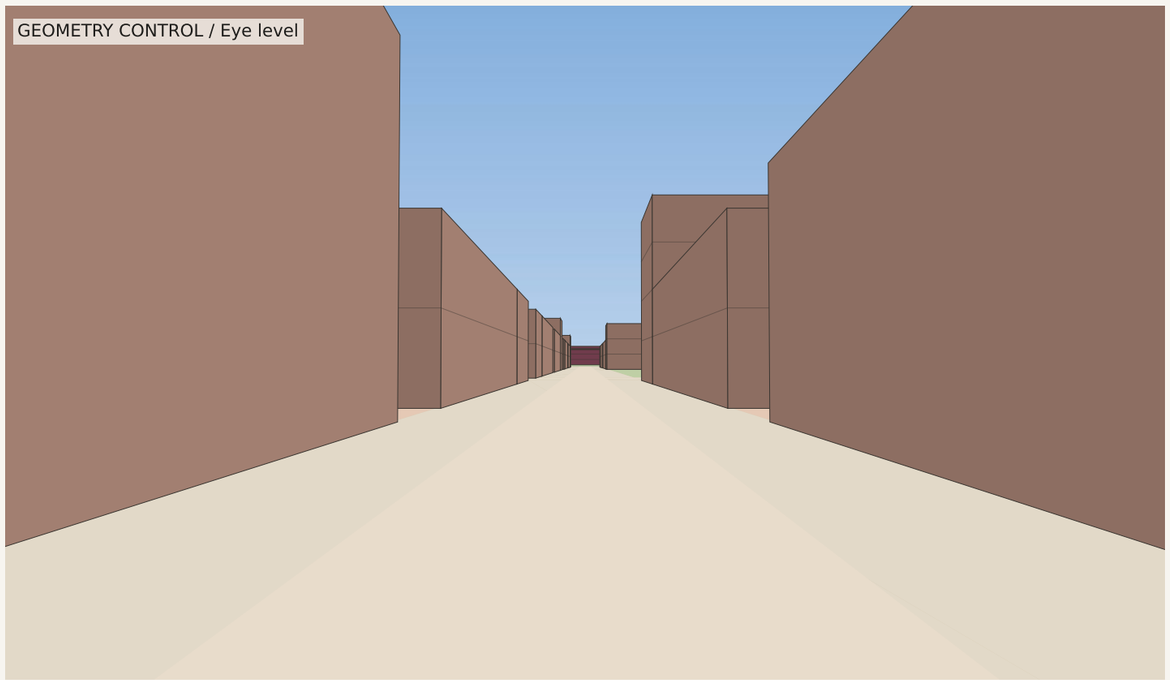


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Terrace Oasis: a city that grows its own shade beside the first fields

Control images remain evidence; concept images illustrate the submitted geometry







A01-aerial · rendered with ChatGPT image generation (chatgpt.com, web app, October 2026) from the massing view A01-aerial · illustrates the geometry, not evidence



A01-eye-day · rendered with ChatGPT image generation (chatgpt.com, web app, October 2026) from the massing view A01-eye-day · illustrates the geometry, not evidence

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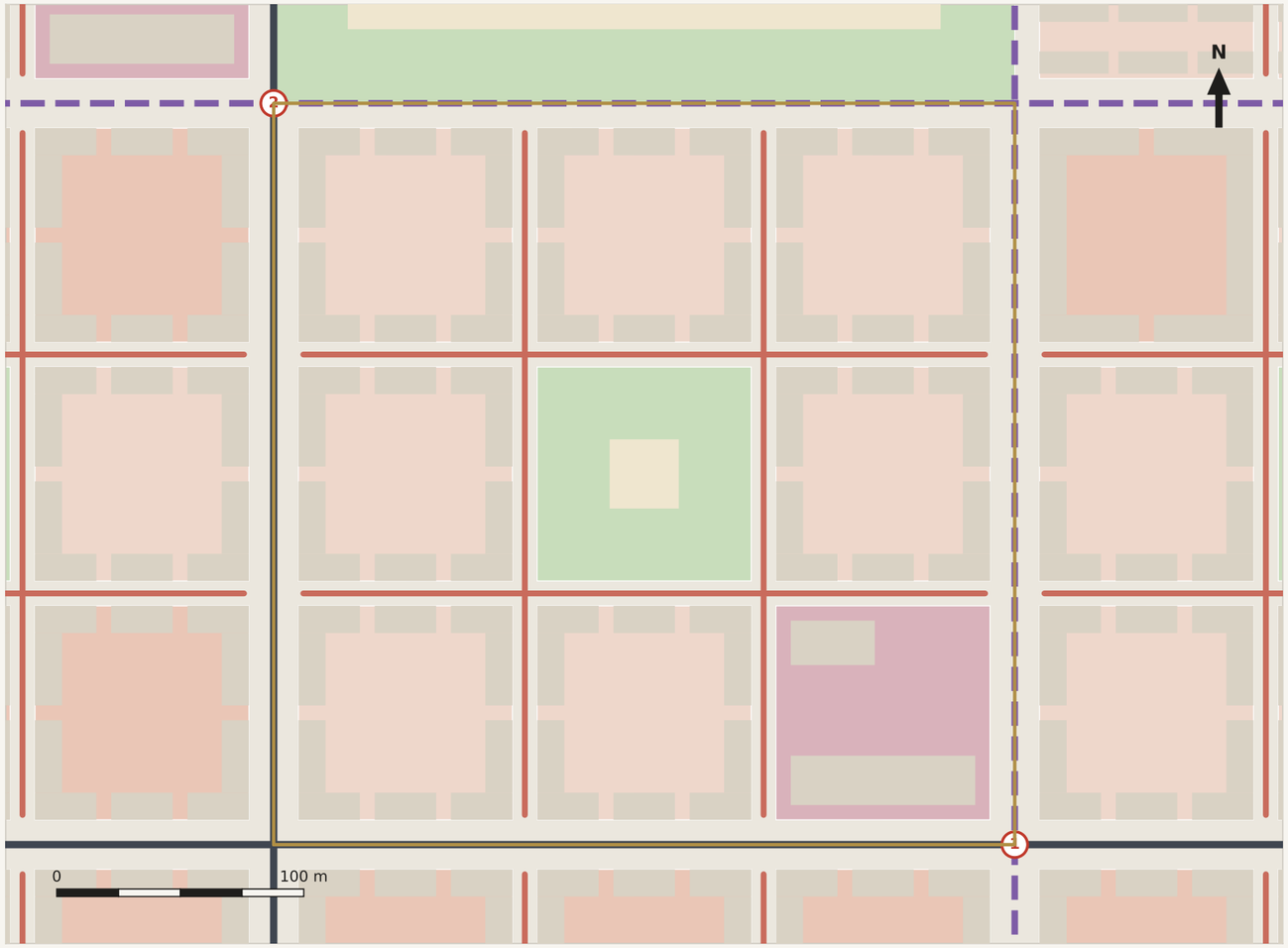
Terrace Oasis: a city that grows its own shade beside the first fields

A day here

	People	Machines
06:00	Windows open to let the night air through the courts; the first walks and the bakery on the corner.	The night's deliveries wait at the drop point on the acequia street; no machine enters a lane.
08:00	Children walk in groups down the lanes to the kindergarten and the school.	Machines on the nearby machine streets are held at the crossings at school time.
13:00	Shutters closed, lunch and siesta indoors; the bottom of each court stays in shade.	Roof PV pergolas run at their peak and charge the block's batteries; robots wait in the shade.
17:00	People who work from home come out to the patio garden; neighbours share tools and food.	Delivery robots bring parcels and groceries from the drop point to the doors at walking pace.
20:00	The patio garden is the cool place at dusk: play, dinners outside, the acequia running.	The block's water is metered and the next day's channel flow is scheduled from it.
Zonda	On a Zonda day families stay in the courts and the lanes, out of the north-west wind.	Air intakes switch to filtered mode; panel cleaning is booked for the day after.

Who and what moves here: networks, interfaces, and the people, robots and shuttles of the 3D scene

People and machines through one day



Interfaces in view

1 Raised table crossing 10
2 Raised table crossing 21

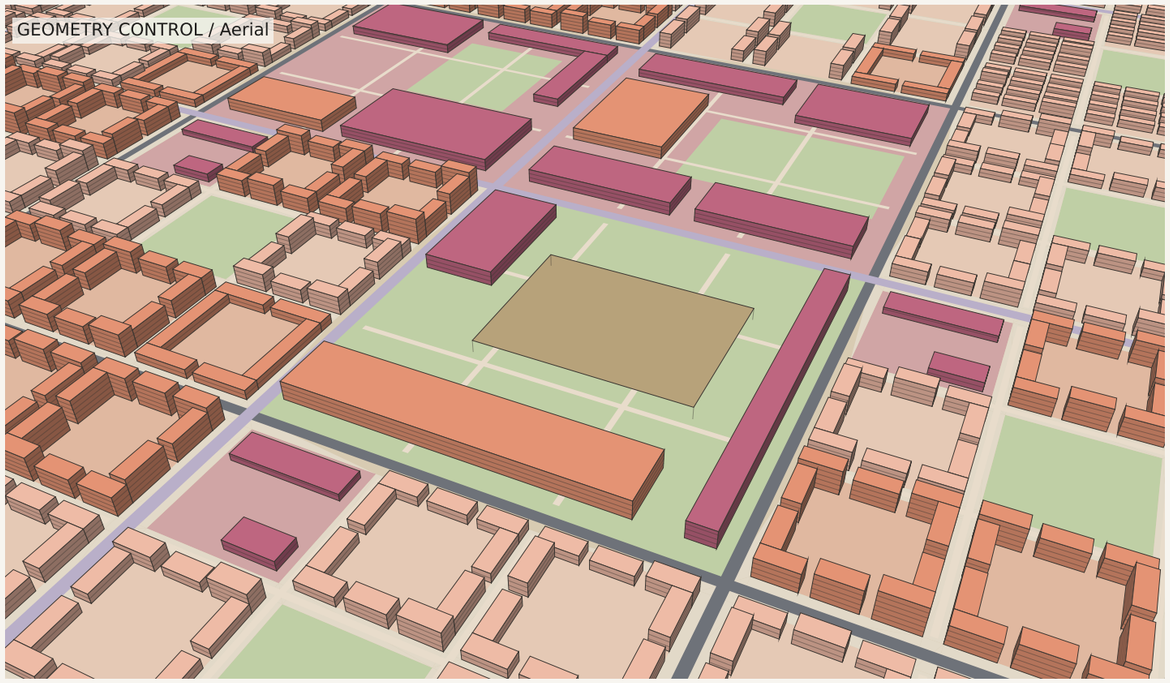
People
Machines + goods
Shared at walking pace
People (scene)
Delivery robots
Driverless shuttles
Human-machine interface

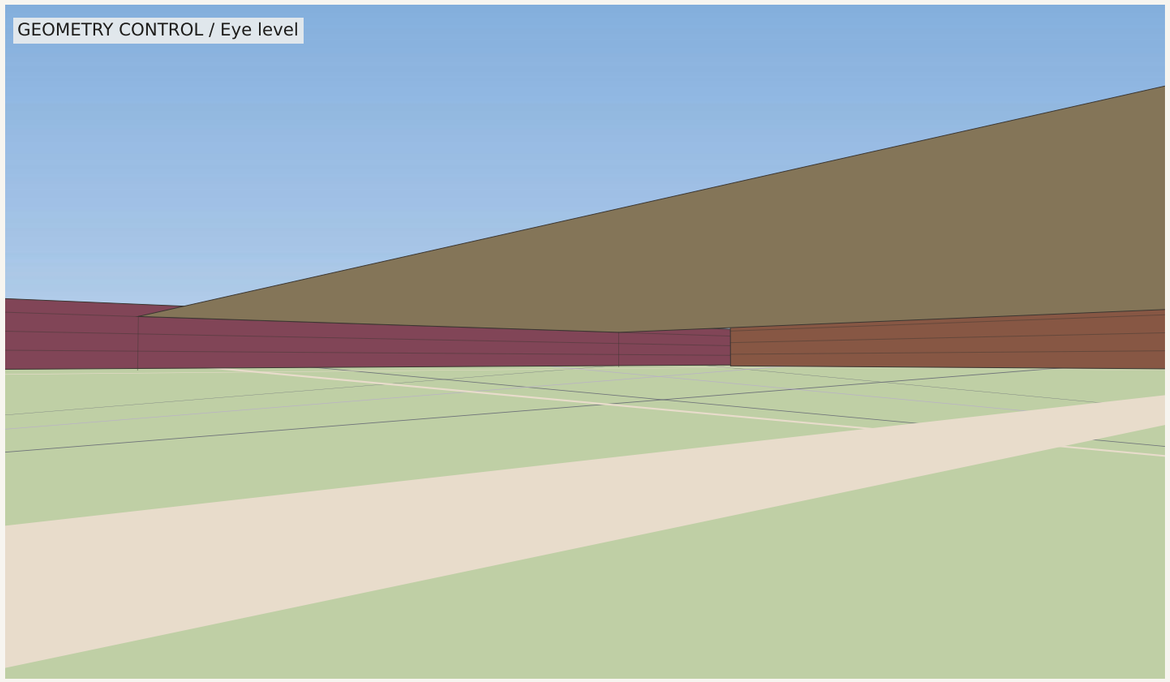


Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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A03-aerial · rendered with ChatGPT image generation (chatgpt.com, web app, October 2026) from the massing view A03-aerial · illustrates the geometry, not evidence



A03-eye-day · rendered with ChatGPT image generation (chatgpt.com, web app, October 2026) from the massing view A03-eye-day · illustrates the geometry, not evidence

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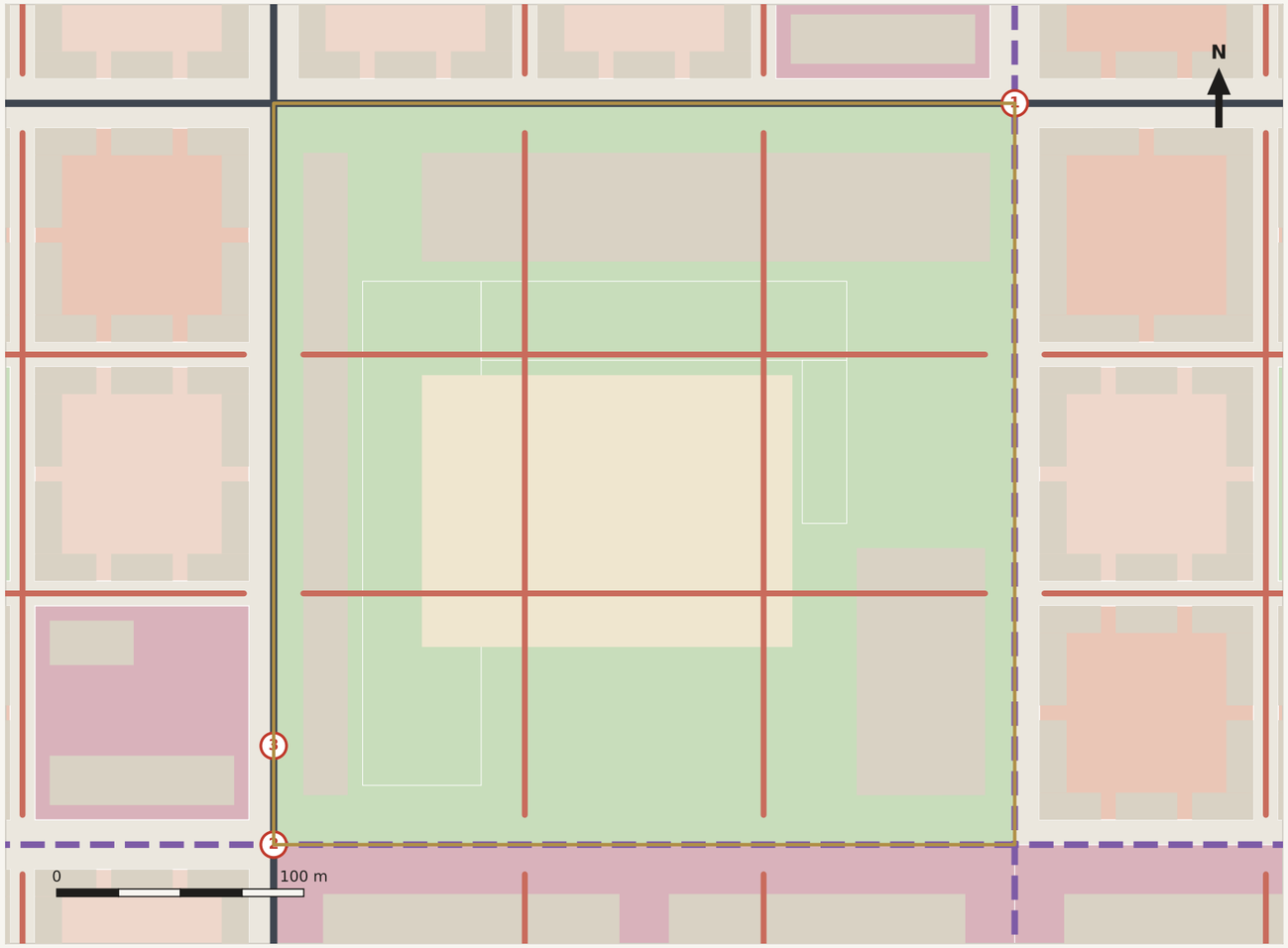
Terrace Oasis: a city that grows its own shade beside the first fields

A day here

	People	Machines
07:00	The market sets up under the canopy; older people walk the square before the heat.	Robots bring produce from the cross-dock to the stalls; the acequia gate opens for the morning.
10:00	Classes and assemblies in the wind gallery; cafes fill along the north arcade.	The canopy's PV feeds the arcade and the cultural hall.
14:00	At 45 C people sit under the canopy and in the grove; the sunken court is the coolest floor in the city.	Water reaches the trees by drip at night, not now: only the basin runs, slowly.
19:00	The square is full: families, music, the market's second shift.	Shuttles stop on the machine street at the north-west corner; robots keep to the shared streets.
23:00	Lights low under the canopy; the last cafes close.	Cleaning robots sweep the square; the channel is flushed to the orchards.
Zonda	In a Zonda people go down into the sunken court or behind the gallery; the market moves inside.	Stall awnings are stowed; the basin is drained to keep dust out of the water.

Who and what moves here: networks, interfaces, and the people, robots and shuttles of the 3D scene

People and machines through one day



Interfaces in view

1 Raised table crossing 6
2 Raised table crossing 18
3 Driverless shuttle stop 1

People
Machines + goods
Shared at walking pace
People (scene)
Delivery robots
Driverless shuttles
Human-machine interface



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Assumptions

A-01 Water source: Alluvial groundwater wellfield up the fan, 2,186,952 m3/yr fresh in phase one (about 69 L/s); 1,412,340 m3/yr more is reclaimed wastewater - No water right is supplied by the organiser and recharge is unknown, so phase two is gated on measured drawdown rather than on a yield we cannot know.

A-02 Phase-one IT load: 40 MW: two 20 MW halls; pads for two more reserved on the terrace for phase two - Sized so that the PV and storage can carry the city first and the campus second, islanded.

A-03 Grid: No grid connection in phase one; the city runs as an islanded microgrid - Grid capacity is not supplied; the nearest road (RN 20) is 40.7 km beyond the site and no transmission line is assumed.

A-04 Cooling: Dry coolers with adiabatic assist above 35 C (about 500 h/yr): WUE 0.20 L/kWh, PUE 1.25 - Water binds, power does not; dry cooling gives up about 0.1 of PUE to save most of the water. WUE per IT kWh.

A-05 PV yield: 2,150 kWh/kWp/yr, single-axis tracking, 2 ha per MWp - Order of magnitude for the high-irradiance Andean piedmont at 31.1 S; to be replaced by a PVGIS run (S-01).

A-06 Storage: 1,000 MWh lithium iron phosphate, round-trip efficiency 88% - Carries the city's night load (about 15 MW) and the campus at 70% for 13 hours with margin.

A-07 Hourly matching and availability: 100% physical hourly carbon-free supply; the city is served first; compute available 85% of hours at full load - Day: 11 h at full load. Night: 13 h at 70% from storage. Winter cloud: batch work curtailed. 11 x 100% + 13 x 70% = 84% of full-load hours.

A-08 Domestic water and reuse: 130 L per person per day; 80% returns as wastewater; 90% of it reused - Metered supply, greywater separation and low-flow fittings in an arid new city; San Juan's own consumption is far higher and is not the benchmark.

A-09 Agricultural, landscape and industrial water: Orchards 226 ha at 4,000 m3/ha/yr deficit drip; greenhouses 14 ha at 8,000; north terrace 338 ha of jojoba and algarrobo at 600 (establishment); irrigated streets, patios and squares 50 ha at 6,000; industry 60,000 m3/yr - Orchards and street trees run on reclaimed water through the acequias; the river park, north slope and PV fields are unirrigated desert planting.

A-10 Dwellings and household size: 85 m2 gross per dwelling; 2.4 persons per dwelling; mixed-use blocks two thirds residential - Courtyard houses and flats of 55-120 m2 for a young city with families.

A-11 Phase-one jobs: 7891 jobs located in the city: Compute campus operations and security 140; PV, batteries and the microgrid 120; Greenhouses 141; Orchards, acequias and landscape crews 420; Agri-processing, cold store and packing 260; Workshops: PV and robot maintenance for the whole site's fields, fabrication, repair 900; Freight cross-dock and autonomous fleet 220; Water reclamation and the bluff tank 60; Schools 790; Kindergartens and health posts 420; Clinic 260; Training college (robotics, agronomy, water) 180; Town hall, open-data office and administration 280; Library, culture, media school 120; Markets, shops, cafes and food 2200; Arrival hall, guesthouse and visitors 260; Personal and home care 700; Lead-zone field operations living in the city 420 - Staffing ratios per facility; the construction workforce of phase two is temporary and excluded.

A-12 Wind: Zonda from the west to north-west, hot and dry, several days a year; ordinary winds not assumed - Wind data is not supplied; to be checked against SMN San Juan Aero observations before phase two.

A-13 Waste-heat demand: Greenhouse night heating about 1.2 GWh/ha/yr; hot water for the civic quarter and the market about 8 GWh/yr: about 25 GWh of about 300 GWh IT energy - The campus is 2.4 km from the nearest homes, so heat goes only to the greenhouses beside it and, by an insulated main down the bluff, to the civic quarter.

A-14 Seismic design: Highest seismic zone of INPRES-CIRSOC 103 assumed; buildings of six floors or fewer, most of two to four; regular, short-span courtyard blocks in reinforced concrete frames or confined masonry - San Juan is among the most seismically active provinces; edition and zone to be confirmed (S-03).

A-15 Long-term size: Four oasis cities of about 40,000 each on the Line, about 160,000 people - Each city stays within a 2.5 km walk across and within reach of its own wellfield; size is limited by water, not land.

A-16 Machine traffic rule: combined: separated on the 600 m machine grid; shared at walking pace on the acequia streets; people only in the lanes - Freight and shuttles never need the lanes; doorstep delivery uses the acequia streets; every crossing of a machine street is a marked interface.

A-17 City energy demand: About 153 GWh/yr for the city (2,000 kWh per resident plus 70 GWh non-residential), 372 GWh/yr for the campus at PUE 1.25 - All-electric city with heat pumps and an electric fleet; to be replaced by an hourly load model.

A-18 Flood: The lower terrace stands 4-15 m above the wash bed and outside the DEM flood-sensitive band; storm water from the bluff is led round the city in two swales to the river park - Flood peaks are not supplied; the city is placed out of the band instead of being raised inside it.

Sources

S-01 Photovoltaic Geographical Information System (PVGIS) - European Commission, Joint Research Centre https://joint-research-centre.ec.europa.eu/photovoltaic-geographical-information-system-pvgis_en

S-02 NASA POWER Data Access Viewer - NASA https://power.larc.nasa.gov/docs/tutorials/data-access-viewer/quick-start/

S-03 INPRES-CIRSOC 103 - Argentine seismic design code - INPRES / CIRSOC

AI disclosure

Agent Claude (Cowork), model claude-opus-5-5, runtime Cloud sandbox, Python 3, this repository's scripts only. Autonomy: human_supervised. The organiser's design lead commissioned a reference example, asked that it answer all three scales as a city rather than a compute town, confirmed the three key nodes the agent proposed (C01, A01, A03), and chose the image tool. The agent made every design decision and wrote every file; no human edited the package.

Images

12 images registered in image_provenance.json. Views in this booklet are, in order of preference: an image made with an image model from a massing view of the geometry (model, prompt and control image registered), the massing view itself (scripts/render_views.py), or a massing drawing by scripts/make_figures.py. All of them explain the geometry; none is evidence.

ChatGPT image generation (chatgpt.com, web app, October 2026) - 12 images, post-processing none. Captured from the browser at 2x device resolution in four tiles and resampled to 1536 x 1024; no retouching. The control image was sent as a 1536 x 960 JPEG of the render; it matches the stamped render within JPEG compression (mean difference under 4/255). Files in media/: C01-aerial.jpg (control C01-aerial.png), C01-eye-day.jpg (control C01-eye-day.png), C01-eye-night.jpg (control C01-eye-night.png), A01-aerial.jpg (control A01-aerial.png), A01-eye-day.jpg (control A01-eye-day.png), A01-eye-evening.jpg (control A01-eye-evening.png), A03-aerial.jpg (control A03-aerial.png), A03-eye-day.jpg (control A03-eye-day.png), A03-eye-evening.jpg (control A03-eye-evening.png), city-aerial.jpg (control city-aerial.png), city-evening.jpg (control city-evening.png), master-aerial.jpg (control master-aerial.png). Full prompts in image_provenance.json.

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